Planar Graph Classes: Visual Guide

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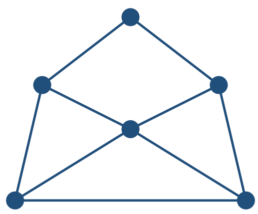
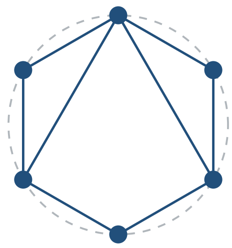
Each picture is a small representative or construction diagram. It is meant to clarify the defining shape; it does not replace the formal Lean definition.

Planar graphs



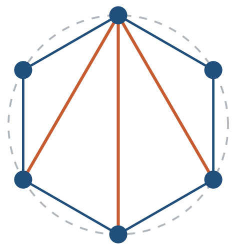
A planar graph admits a drawing with no two edges crossing away from their common endpoints.

Outerplanar graphs



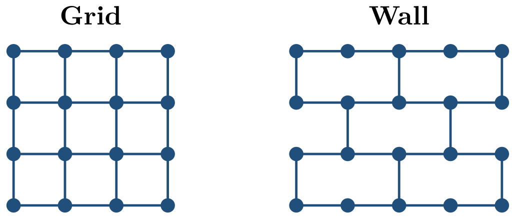
Every vertex lies on the boundary of the outer face; the dashed circle emphasizes the common boundary.

Maximal outerplanar graphs



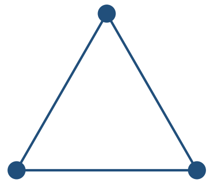
The boundary polygon is fully triangulated by noncrossing diagonals. Adding any further edge destroys outerplanarity.

Grids and walls



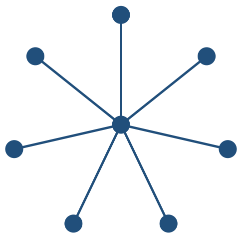
A grid keeps every horizontal and vertical neighbor edge. A wall keeps alternating vertical edges, producing the brick pattern.

Triangles



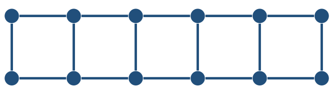
A triangle is the complete graph on three vertices: every pair of vertices is adjacent.

Stars



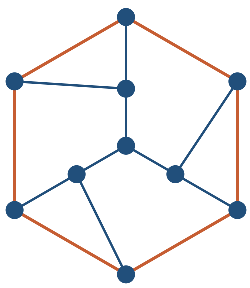
One center is adjacent to every leaf, while no two leaves are adjacent.

Ladders



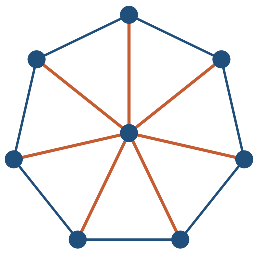
Two paths form the rails, and corresponding vertices are joined by rungs.

Halin graphs



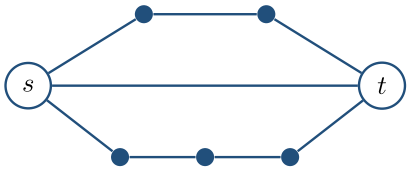
Start with a plane tree having no degree-two vertex, then join its leaves in cyclic order to form the highlighted rim.

Wheels



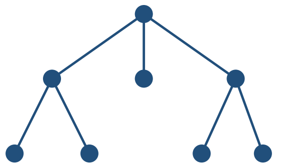
A rim cycle surrounds one universal hub, which is adjacent to every rim vertex.

Series-parallel graphs



Series composition subdivides a terminal route; parallel composition places terminal routes side by side between s and t.

Trees

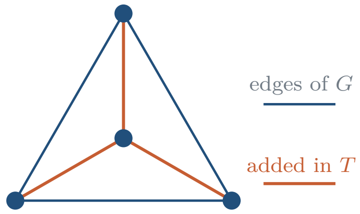


A tree is connected and contains no cycle; consequently there is a unique path between every pair of vertices.

Paths

A path is a linear chain: two endpoints have degree one and every internal vertex has degree two.

Triangulations



A triangulation adds edges to a planar graph on the same vertex set until no further edge can be added without losing planarity. In a plane embedding, every face is then triangular.